

Integration of Classy-Blocks and OpenFOAM within Blender

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Weekly Progress Report

OpenFOAM GUI

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March 21, 2026

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Introduction

- ◀ **OpenFOAM:** A C++ toolbox for CFD using finite volume methods.
- ◀ **Classy-Blocks:** A Python library for procedural `blockMeshDict` generation, mimicking modern 3D CAD modeling.
- ◀ **Blender Integration:** Utilizing Blender as a frontend for geometry extraction and mesh visualization using `pyvista`.

Research Gap

- ◀ **Gap 1:** Lack of native Blender 5.x support for advanced Python meshing libraries like `classy_blocks`.
- ◀ **Gap 2:** Absence of an integrated “One-Click” solution for geometry extraction, mesh building, and quality reporting within a unified environment.

Problem Statement

- ◀ To develop a Blender Add-on that leverages `classy_blocks` and `pyvista` to automate OpenFOAM case setup, allowing users to extract geometry, generate high-quality meshes, and execute solvers without leaving the Blender interface.

Work Done: Global Settings & Panel Structure

- ◀ **Addon Preferences Panel:** Cleaned up the 3D viewport by offloading system-level parameters (e.g., bashrc paths, default case paths, and solver tolerances) into the native Blender Edit → Preferences registry.
- ◀ **UI Structural Layout:** Reorganized the custom operator properties into visually distinct, grouped layouts (`layout.box()`). Integrated conditional visibility to hide irrelevant fields, such as hiding STL projections unless the Box block type is selected.

Work Done: Event Tracking & Quality Visuals

- ▶ **Diagnostics HUD:** Engineered a dynamic, timestamped Heads-Up Display property (`pipeline_status`) at the top of the add-on panel, tracking multi-stage OpenFOAM processes in real-time without freezing the UI narrative.
- ▶ **Visual Quality Formatting:** OpenFOAM mesh metrics are now visually evaluated against solver tolerances. Non-orthogonality and Skewness string values automatically map to Red/Amber/Green iconography for immediate visual user feedback.
- ▶ **Safe Active Contexts:** Added an automated context check for the Terrain STL export functionality, utilizing Blender's active selection states to securely grey-out operators if an export is impossible.